

MTH201 RHEMA QUESTIONS WITH Engr. aLmighty Formula

1. If $y = 2x^3 + \frac{1}{x^2} + 4$. Find dy/dx a. $6x^2 + \frac{1}{2x^2}$ b. $6x^2 - \frac{1}{2x}$ c. $6x^2 - \frac{2}{x}$ d. $6x^2 - \frac{2}{x^3}$
2. If $y = 7x^2 - \sqrt{x}$ Find d^2y/dx^2 a. $14x - \frac{\sqrt{x^3}}{2}$ b. $14x - \frac{1}{2x}$ c. $14 + \frac{1}{4\sqrt{x^3}}$ d. $14 - \frac{1}{4\sqrt{x^3}}$
3. Differentiate $(2x + 4)^5$ at $x = -2$ a. 0 b. 8 c. -1 d. NOTA
4. Differentiate $\sin^4 x$ a. $4\sin^3 x$ b. $4\sin^3 x \cos x$ c. $\cos^4 x$ d. $4\cos^3 x$
5. Differentiate $\sqrt{x^2 + \tan x}$ a. $\sqrt{2x + \sec^2 x}$ b. $\frac{1}{\sqrt{2x^2 + \tan x}}$ c. $\frac{2x + \sec^2 x}{2\sqrt{x^2 + \tan x}}$ d. NOTA
6. Differentiate $\sin(x^2 + e^x)$ a. $\cos(x^2 + e^x)$ b. $\cos(2x + e^x)$ c. $(2x + e^x)\sin(x^2 + e^x)$ d. $(2x + e^x)\cos(x^2 + e^x)$
7. Find the first derivative of the function $e^{x^2 + \sin^2 x + 1}$ a. $e^{x^2 + \sin^2 x + 1}$ b. $(2x + \cos^2 x + 1)e^x$ c. $(2x + \cos^2 x + 1)e^{x^2 + \sin^2 x + 1}$ d. NOTA
8. Differentiate $x^4 e^x$ a. $x^4 e^x + 4x^3 e^x$ b. $(x^4 + 4x^3)e^x$ c. $x^3 e^x(x + 4)$ d. All of the above
9. Differentiate $\frac{x^2 + e^x}{\sin x}$ a. $\frac{2x + e^x}{\cos x}$ b. $\frac{2x + e^x}{\sin x}$ c. $\frac{x^2 + e^x}{\cos x}$ d. NOTA
10. $x^3 y + 2x \cos y + y \tan x = 4$. Find dy/dx a. $\frac{-(3x^2 y + 2 \cos y + y \sec^2 x)}{2x^3 y - 2x \sin y + \tan x}$ b. $\frac{3x^2 y + 2 \cos y + y \sec^2 x}{2x^3 y - 2x \sin y + \tan x}$ c. $\frac{-(2x^3 y - 2x \sin y + \tan x)}{3x^2 y + 2 \cos y + y \sec^2 x}$ d. $\frac{(2x^3 y - 2x \sin y + \tan x)}{3x^2 y + 2 \cos y + y \sec^2 x}$
11. Find the n th differential coefficient of $y = \ln x$ a. $(-1)^{nn!} x^{-n}$ b. $(-1)^{n-1} (n-1)! x^{-n}$ c. $n! x^{-n}$ d. NOTA
12. If $f(x) = e^{ax}$. Find the n th derivative of $f(x)$. a. e^{ax} b. $2e^{ax}$ c. $2^n e^{ax}$ d. NOTA
13. If $f(x) = (x^2 + 1)e^{2x}$, find $f^n(x)$ where $f^n(x)$ is the n th derivative of $f(x)$ a. $2^{n-2} e^{2x} [4x^2 + 4nx + n^2 - n + 4]$ b. $2^{n+2} e^{2x} [4x^2 + 4nx + n^2 - n + 4]$ c. $2^{n-2} e^{2x} [4x^2 - 4nx + n^2 - n + 4]$ d. $2^{n-2} e^{2x} [4x^2 - 4nx + n^2 - n - 4]$ e. NOTA
14. If $f(x) = \cos x$, expand around $x = 0$ a. $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \dots$ b. $1 - \frac{x^2}{2!} - \frac{x^4}{4!} - \frac{x^6}{6!} \dots$ c. $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} \dots$ d. $1 + \frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} \dots$
15. Expand $\ln(1 + \sin x)$ around $x = 0$. a. $x - \frac{x^2}{2!} + \frac{2x^3}{3!} \dots$ b. $x + \frac{x^2}{2!} + \frac{2x^3}{3!} \dots$ c. $x - \frac{x^2}{2!} - \frac{2x^3}{3!} \dots$ d. NOTA
16. Which of this represents the expansion of $f(x)$ centered around $x = a$ a. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-b)^n}{n!}$ b. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x+b)^n}{n!}$ c. $f(x) = \sum_{n=0}^{\infty} \frac{f(a)(x-b)^n}{n!}$ d. NOTA
17. Find the first 4 terms in the Taylor series for $(x - 1)e^x$ near $x = 1$ a. $e(x - 1) + e(x - 1)^2 + \frac{e(x-1)^3}{2} \dots$ b. $e(x - 1) + \frac{e(x-1)^2}{2} + \frac{e(x-1)^3}{2} \dots$ c. $(x - 1)(x - 1)^2 \frac{(x-1)^3}{2} \dots$ d. NOTA
18. Use Maclaurin Series to Expand $x \cos x$ a. $x - \frac{x^3}{2} + \frac{x^5}{24} \dots$ b. $x + \frac{x^3}{2} + \frac{x^5}{24} \dots$ c. $x - \frac{x^2}{2} + \frac{x^3}{6} \dots$ d. $x + \frac{x^2}{2} - \frac{x^3}{6} \dots$
19. Expand $x \sin x$ using Taylor series if centered at $x = \pi/2$ a. $\pi/2 + (x - \pi/2) - \frac{\pi(x - \pi/2)^2}{4} - \frac{(x - \pi/2)^3}{2} \dots$ b. $\pi/2 - (x - \pi/2) - \frac{\pi(x - \pi/2)^2}{4} - \frac{(x - \pi/2)^3}{2} \dots$ c. $\frac{\pi}{2} + \left(x - \frac{\pi}{2}\right) + \frac{\pi(x - \pi/2)^2}{4} - \frac{(x - \pi/2)^3}{2} \dots$ d. NOTA
20. Expand e^x a. $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} \dots$ b. $1 + x - \frac{x^2}{2!} + \frac{x^3}{3!} \dots$ c. $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \dots$ d. $1 - x - \frac{x^2}{2!} - \frac{x^3}{3!} \dots$

21. Expand $\sin x$ using Maclaurin Series a. $x + \frac{x^2}{2!} + \frac{x^3}{3!} \dots$ b. $x - \frac{x^2}{2!} + \frac{x^3}{3!} \dots$ c. $x - \frac{x^2}{2!} - \frac{x^3}{3!} \dots$ d. NOTA.
22. Expand xe^x a. $\sum_{n=1}^{\infty} \frac{x^n}{n!}$ b. $\sum_{n=1}^{\infty} \frac{(xe^x + ne^x)x^n}{(n-1)!}$ c. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ d. $\sum_{n=1}^{\infty} \frac{(xe^x + ne^x)x^n}{n!}$ e. NOTA
23. Find the Taylor series expansion of $\ln x$ centered at $x = 1$. A. $(x - 1) - \frac{(x - 1)^2}{2} + \frac{(x - 1)^3}{3} \dots$ B. $(x - 1) + \frac{(x - 1)^2}{2} + \frac{(x - 1)^3}{3}$ c. $(x - 1) - \frac{(x - 1)^2}{2} - \frac{(x - 1)^3}{3}$ d. NOTA
24. Find all numbers C that satisfy the mean value theorem if it can be applied to $f(x) = x^2 - 4x + 1$ on the closed interval $[1, 5]$. a. 1 b. 2 c. 3 d. NOTA
25. Find all numbers c that satisfy the mean value theorem if it can be applied to $f(x) = x^{\frac{2}{3}}$ on the interval $[0, 1]$ a. 1 b. 2 c. 3 d. 4 e. NOTA
26. Determine the mean value theorem for the given function and interval $h(z) = 4z^3 - 8z^2 + 7z - 2$ on the interval $[2, 5]$ a. 107 b. 108 c. 109 d. NOTA
27. Determine the mean value theorem for the given function and interval $\Delta(t) = 8t + e^{-3t}$ on the interval $[-2, 3]$ a. -363.4 b. 363.4 c. 491.2 d. 419.3 e. NOTA
28. Determine the values of c for the function $f(x) = \sqrt{x - 3}$ on the interval $[3, 7]$ a. 1 b. 2 c. 3 d. 4 e. NOTA
29. Let $f(x) = \sqrt{4x - 3}$ and let c be the number that satisfies the mean value theorem for f on the interval $1 \leq x \leq 3$. a. 1.75 b. 1.25 c. 1.5 d. NOTA
30. Find the linear approximation of the $f(x) = (1 + x)^4$ at $x = 0$ use this approximation to estimate $(1.01)^4$ a. 1.01 b. 1.02 c. 1.03 d. 1.04
31. Find the linear approximation for $f(x) = \cos x$ at $x = \frac{\pi}{2}$ use this to approximate $\cos 92^\circ$ a. -.86 b. -0.034 c. -0.42 d. -0.911
32. Find the linear approximation for $f(x) = \sin x$ at $x = \frac{\pi}{3}$ use this to approximate $\cos 62^\circ$ a. -.86 b. -0.034 c. -0.42 d. -0.911 e. NOTA
33. Find the local linear approximation of $f(x) = 3\sqrt{x}$ at $x = 8$. Use it to approximate $3\sqrt{8}$
34. Find the linearization $L(x)$ of the function $f(x) = \sqrt{x}$ at $x = 36$. Use it to estimate $f(37)$ a. 6.083 b. 6.012 c. 6.037 d. NOTA
35. Given that $f(x) = x^3$, find the linearization $L(x)$ of the function at $x = 2$. Use it to estimate $f(2.1)$ a. 9.2 b. 8.2 c. 8.9 d. 10 e. NOTA
36. Estimate the given number using a local linear approximation a. 2.0833 b. 2.8333 c. 2.0083 d. NOTA
37. Estimate $\ln 1.05$ using linear approximation. A. 0.05 b. 0.04 d. 0.03
38. Estimate $\tan 46^\circ$ using Linear approximation. a. 1.03553 b. 1.034907 c. 1 d. NOTA
39. Find the stationary points to the surface $z = x^3 + xy + y^2$ a. $(0, 0), (1/6, -1/12)$ b. $(1, 1)$ $(1/6, -1/12)$ c. $(0, 0), (1/6, 1/12)$ d. All of the above e. NOTA
40. A function $f(x, y)$ is said to have relative maximum at point (x_0, y_0) if which of the following condition holds a. $f_{xx} < 0$ and $f_{xx} f_{yy} > (f_{xy})^2$ b. $f_{xx} > 0$ and $f_{xx} f_{yy} > (f_{xy})^2$ c. $f_{xx} f_{yy} < (f_{xy})^2$ d. $f_{xx} f_{yy} > (f_{xy})^2$ e. NOTA
41. A function $f(x, y)$ is said to have a saddle point at point (x_0, y_0) if which of the following condition holds a. $f_{xx} f_{yy} < (f_{xy})^2$ b. $f_{xx} f_{yy} > (f_{xy})^2$ c. $f_{xx} < f_{xy}$ d. $f_{xx} f_{yy} > (f_{xy})^2$ e. NOTA

42. Find the minimum and maximum values of $f(x,y) = xy$ subject to the constraint $x^2 + y^2 = 8$ a. (8,4) b. (8, -1) c. (4,-4) d. (2,1) e. NOTA
43. Find the smallest value of the function $f(x,y) = x^2 + y^2$ subject to the constraint $y + 3x = 3$ a. (0.9, 0.3) b. (0.9, -0.3) c. (-0.9, 0.3) d. (-0.9, -0.3) e. NOTA
44. Find the value of the Lagrange multiplier for $f(x,y,z) = x + y + 2z$ subject to the constraint $x^2 + y^2 + z^2 = 3$ a. $\pm \frac{1}{2}$ b. $\pm \frac{1}{4}$ c. $\pm 1/\sqrt{2}$ d. NOTA
45. Find the values of the langrage multiplier that gives the smallest value of the function $(x,y) = x^2 + y^2$ subject to the constraint $y + 3x = 3$ a. 1 b. 2 c. 0.2 d. 0.6 e. NOTA
46. Given $f(x,y,z) = 3x^2 + y^2 - 2x^2$ and $3x + 2y - 8z = -50$. Use Lagrange multipliers to find any maximum or minimum values . a. 100 b. -100 c. 50 d. -50
47. Given $f(x,y,z) = 4x+2y+6y$ and $x^2+y^2+z^2=14$ find the Lagrange multiplier of the above function. a. $\pm \frac{1}{2}$ b. $\pm \frac{1}{4}$ c. ± 1 d. NOTA
48. Find the Lagrange multipliers of the function $f(x,y,z) = xyz$ and the constraint $x^2+4y^2+16z^2 = 48$ a. a. $\pm \frac{1}{2}$ b. ± 2 c. ± 1 d. NOTA
49. If $f(x, y, z) = xy + z$; find $\frac{\partial f}{\partial x}$ a. y b. x c. z d. xy e. NOTA
50. If $f(x, y, z) = xy + z$; find $\frac{\partial f}{\partial z}$ a. y b. x c. z d. xy e. NOTA
51. Given that $x = r\cos\theta$ and $y = r\sin\theta$ find ds if $ds^2 = dx^2 + dy^2$ a. $\sqrt{dr^2 + r^2 d\theta^2}$ b. $\sqrt{dr^2 - r^2 d\theta^2}$ c. $\sqrt{d\theta^2 + r^2 dr^2}$ d. 0 e. NOTA
52. Given $U(x, y) = e^x \cos y$. Find $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}$ a. 0 b. 1 c. 2 d. -1 e. NOTA
53. Given $U(x, y, z) = x^2 + 2xy - y \ln z$ and $x = s + t^2, y = s - t^2, z = 2t$ find $\frac{\partial U}{\partial s}$ a. $6s + 2t^2 - \ln 2t$ b. $6 + 2t^2 - \ln 2t$ c. $6s + 2t - \ln 2t^2$ d. $6s^2 + 2t^2 + \ln 2t$ e. NOTA
54. Given that $y = r\sin\theta$, find dy a. $\sin\theta dr + r\cos\theta d\theta$ b. $\sin\theta dr - r\cos\theta d\theta$ c. $\sin^2\theta dr + r^2\cos\theta d\theta$ d. $\sin\theta d\theta + r\cos\theta dr$ e. NOTA
55. Differentiate $f(x, y, z) = e^{1-x\cos y} + ze^{\frac{-1}{(1+y^2)}}$ once with respect to x at $(1,0, \pi)$ gives a. 1 b. -1 c. -2 d. 0 e. NOTA
56. The partial derivative of $f(x,y)$ with respect to y is given as a. $\lim_{h \rightarrow 0} \frac{f(x+h, y)-f(x,y)}{h}$ b. $\lim_{h \rightarrow 0} \frac{f(x+h, y)+f(x,y)}{h}$ c. $\lim_{h \rightarrow 0} \frac{f(x, y+h)-f(x,y)}{h}$ d. $\lim_{h \rightarrow 0} \frac{f(x, y+h)+f(x,y)}{h}$
57. Find the first partial derivative of $f(x, y) = x^3y^2 + 3xe^{-y}$ with respect to y. a. $3x^2y^2 + 3e^{-y}$ b. $2x^3y - 3xe^{-y}$ c. $2x^3y + 3xe^{-y}$ d. $3x^2 + 2xe^{-y}$ e. NOTA
58. Find f_y when $f(x,y) = y\sin(xy)$. a. $\sin(xy) + \cos(xy)$ b. $\sin(xy) + xycos(xy)$ c. $\sin(xy) + y$ d. $\cos(xy)$ e. NOTA
59. Find f_{yx} for $f(x,y) = x\sin(xy)$ a. $2xcos(xy) - x^2ysin(xy)$ b. $2xcos(xy) + x^2ysin(xy)$ c. $2xsin(xy) - x^2ycos(xy)$ d. $2xsin(xy) + x^2ycos(xy)$ e. NOTA
60. Find f_{xxyy} if $f(x,y) = x^4y^4$ a. $64 x^4y^4$ b. $144x^4y^4$ c. $144x^2y^2$ d. $144x^2y^2$ e. NOTA
61. A vibrating string has a displacement $y = U(x, t)$ in cm at a distance x in cm from one end and at time t in seconds where $U(x, t) = 2\sin(120\pi(x - t))$. How fast (in units of cm per seconds) is the string vibrating at a horizontal distance $x = 1.2\text{cm}$ from one end at time $t = 2\text{s}$. a. 120π b. 240π c. 280π d. NOTA

62. Suppose that a string is attached at its endpoints $x = 0$ and $x = l$, for some number l . Suppose also that $y = U(x, t)$ models the displacement at x in $(0, l)$ of the string at time t , where $U(x, t) = A \cos(at) \sin\left(\frac{\pi}{l}x\right)$ with A and a a constant. What is the rate of change of $u(x, t)$ at $x = l/2$ at any given time? a. 0 b. 1 c. $Aa \sin(at)$ d. $-Aa \sin(at)$ e. NOTA
63. The function $U(x, t) = e^{-t} \sin^2(\pi x) + 32$ models the temperature in $^{\circ}\text{F}$ of a 1 foot long thin rod in which both ends are held at the freezing point at all times t . How fast is the temperature decreasing at the midpoint of the rod when $t = 0$? a. 0 b. 1 c. e d. NOTA
64. The total resistance R produced by two resistors with resistances R_1 and R_2 respectively satisfies $R = \frac{R_1 R_2}{R_1 + R_2}$. What is the rate of change of total resistance R with respect to the resistance R_1 a. $\frac{R_1^2}{(R_1 + R_2)^2}$ b. $\frac{R_2^2}{(R_1 + R_2)^2}$ c. $\frac{R_1^2}{(R_1 + R_2)^2}$ d. $\frac{R_2^2}{(R_1 + R_2)^2}$ e. NOTA
65. Given that $y = x^3 + 2x^2$ find the value of dy if $dx = 0.1$ and $x = 1$ a. 0.1 b. 0.2 c. 0.3 d. 0.4
66. Evaluate $\int (1 - 4x)^3 dx$ a. $\frac{(1-4x)^4}{4} + C$ b. $\frac{-(1-4x)^4}{4} + C$ c. $\frac{(1-4x)^4}{16} + C$ d. $\frac{-(1-4x)^4}{16} + C$
67. Evaluate $\int_0^{3.5} \sqrt[3]{2x+1} dx$ a. 5.125 b. 5.625 c. 6.525 d. 6.125
68. Evaluate $\int \frac{4x}{3-x^2} dx$ a. $2 \log_e |3 - x^2| + C$ b. $4 \log_e |3 - x^2| + C$ c. $-4 \log_e |3 - x^2| + C$ d. $-2 \log_e |3 - x^2| + C$ e. NOTA
69. Solve $\int \tan^3 x \sec^2 x dx$ a. $\frac{\tan^4 x}{4} + \frac{\sec^4 x}{3} + C$ b. $\frac{\tan^4 x}{4} - \frac{\tan^2 x}{2} + C$ c. $\frac{\sec^4 x}{4} - \frac{\sec^2 x}{2} + C$ d. $\frac{\tan^4 x}{4} - \frac{\sec^3 x}{3} + C$ e. NOTA
70. Solve $\int \sin^3 x \cos^2 x dx$ a. $\frac{\sin^2 x}{4} + \frac{\sin^6 x}{6} + C$ b. $-\frac{\cos^4 x}{4} - \frac{\cos^6 x}{6} + C$ c. $\frac{\sin^4 x}{4} - \frac{\sin^6 x}{6} + C$ d. $\frac{\cos^4 x}{4} - \frac{\cos^6 x}{6} + C$ e. NOTA
71. Solve $\int \sin x \cos x dx$ a. $\frac{\sin^2 x}{2} + C$ b. $\frac{\sin^2 x \cos x}{2} + C$ c. $\frac{\cos^2 x \sin x}{2} + C$ d. $\cot x + C$
72. Evaluate $\int x^2 \sin x dx$ a. $\sin x(2 - x^2) + x \cos x + C$ b. $\cos x(2 - x^2) + x \sin x + C$ c. $\sin x(2 + x^2) - x \cos x + C$ d. $\sin x(2 - x^2) + x \sin x + C$ e. NOTA
73. Evaluate $\int x e^x dx$ a. $(x - 1) e^x + C$ b. $(x + 1) e^x + C$ c. $\frac{x^2 e^x}{2} + x + C$ d. $x^2 e^x + C$ e. NOTA
74. Evaluate $\int x^2 \ln x dx$ a. $\frac{x^3}{3} (\ln x + 1) + C$ b. $\frac{x^3}{27} (3 \ln x + 1) + C$ c. $\frac{x^3}{9} (3 \ln x - 1) + C$ d. $x^3 (3 \ln x - 1) + C$ e. NOTA
75. Solve $\int_0^{1/2} 4x e^{2x} dx$ a. $2 - e$ b. $2e$ c. 2 d. $2 + e$ e. NOTA
76. Evaluate $\int_0^1 x^2 e^{-x} dx$ a. $2 - 5e^{-1}$ b. $2(1 - e^{-1})$ c. $2 + 5e^{-1}$ d. NOTA
77. Evaluate $\int (\cos^3 x + 2) \sin x dx$ a. $\left(\frac{\cos^4 x}{4} - 2 \cos x\right) + C$ b. $-\left(\frac{\cos^4 x}{4} + 2 \cos x\right) + C$ c. $-\left(\frac{2 \sin x}{4} - 2 \cos^4 x\right) + C$ d. $\left(\frac{\sin^4 x}{4} - 2 \sin x\right) + C$ e. NOTA
78. Solve $\int x \sqrt{x+1} dx$ a. $\frac{2(x+1)^{2/5}}{5} - \frac{2(x+1)^{2/3}}{3} + C$ b. $\frac{(x+1)^{2/5}}{10} + \frac{(x+1)^{2/3}}{6} + C$ c. $\frac{x(x+1)^{2/5}}{5} + \frac{(x+1)^{2/3}}{3} + C$ d. $\frac{(x+1)^{2/5}}{5} - \frac{(x+1)^{2/3}}{3} + C$ e. NOTA
79. Solve $\int x(x^2 + 2)^3 dx$ a. $\frac{(x^2+2)^4}{4} + C$ b. $\frac{x(x^2+2)^4}{8} + C$ c. $\frac{(x^2+2)^4}{8} + C$ d. $\frac{x(x^2+2)^4}{16} + C$ e. NOTA

80. $\int \frac{x}{\sqrt{1+2x^2}} dx$ a. $\frac{(1+2x^2)^{-1/2}}{2} + C$ b. $\frac{(1+2x^2)^{1/2}}{2} + C$ c. $\frac{x(1+2x^2)^{1/2}}{2} + C$ d. $x(1+2x^2)^{1/2} + C$
81. Evaluate $\int 2^{\sec x} \sec x \tan x dx$ a. $\frac{2^{\sec x}}{\ln 2} + C$ b. $\frac{2^{\tan x}}{\ln 2} + C$ c. $2^{\sec x \tan x} \ln 2 + C$ d. $2^{\sec x \tan x} + C$
e. NOTA
82. If $\int \frac{x dx}{x^3 - 7x - 6} = A \ln|x+1| + B \ln|x+2| + C \ln|x-3| + k$ a. $\frac{1}{4}, -2/5, 3/20$ b. $-1, 2, 3$ c. $\frac{1}{2}, -2, 3/20$ d. $-1, -2/5, 3$ e. NOTA
83. Solve $\int \frac{dx}{9-x^2}$ a. $\frac{1}{6} \ln\left(\frac{3-x}{3+x}\right) + C$ b. $\frac{1}{6} \ln\left(\frac{3+x}{3-x}\right) + C$ c. $\frac{1}{3} \ln\left(\frac{3-x}{3+x}\right) + C$ d. $\frac{1}{3} \ln\left(\frac{3+x}{3-x}\right) + C$ e. NOTA
84. Evaluate $\int \sqrt{\frac{x}{2-x}} dx$ a. 0.658 b. 0.527 c. 9.231 d. NOTA
85. Solve $\int \arctan x dx$ a. $\arctan x - \ln(1+x^2) + C$ b. $x \arctan x - 2 \ln(1+x^2) + C$ c. $\arctan x - \frac{1}{2} \ln(1+x^2) + C$ d. $\arctan x - x \ln(1-x^2)$ e. NOTA
86. The line integral of the function $f(x,y)$ ranging from $t=a$ to $t=b$ is given as where x and y are functions of t a. $\int_a^b f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ b. $\int_a^b f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2} dt$ c. $\int_a^b f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^3 + \left(\frac{dy}{dt}\right)^3} dt$ d. NOTA
87. C is given by $x = 4t$ and $y = 3t$, where t runs from 0 to 2. Evaluate $\int_C \frac{y^2}{x} dS$ a. 22.5 b. -22.5
c. 45 d. -45
88. C is given by $x=t^2, y=t^3, z=t^2$, where t runs from 0 to 1. Evaluate $\int_C (zdx + xdy + ydz)$
a. $33/20$ b. $-33/20$ c. $45/2$ d. $-45/2$
89. Evaluate $\int_C f dS$ with $f(x,y,z) = x^2 + y - 4z$ over C the line segment from $(4,3,1)$ to $(0,1,1)$
a. $\frac{31\sqrt{5}}{3}$ b. $\frac{32\sqrt{5}}{3}$ c. $\frac{32\sqrt{5}}{3}$ d. NOTA
90. Evaluate $I = \int (x+y) dx$ from $H(0,1)$ to $B(0,-1)$ along the semi-circle $x^2+y^2=1$ for $x \geq 0$
91. Evaluate $\int_0^1 \int_0^x (x^2 + y^2) dA$, where dA indicates small area in xy -plane. a. $\frac{1}{2}$ b. $\frac{1}{3}$ c. $\frac{1}{4}$
d. $\frac{2}{3}$
92. $\int_R (x+y) dy dx$, R is the region bounded by $x=0, x=2, y=x, y=x+2$ a. 12 b. 13 c. 14
d. 15
93. $\int_R xy dx dy$, where R is the quadrant of the circle $x^2 + y^2 = a^2$ where $x \geq 0$ and $y \geq 0$ a. $\frac{a^4}{2}$ b. $\frac{a^4}{4}$ c. $\frac{a^4}{6}$ d. $\frac{a^4}{8}$ e. NOTA
94. Evaluate $\int_S \sqrt{xy - y^2} dy dx$, where S is a triangle with vertices $(0,0), (10,1)$ and $(1,1)$ a. 2 b. 4 c. 6 d. 8
95. Find the area bounded by $y = x - 2$ and $y^2 = 2x + 4$ a. 9 b. 18 c. 22 d. 13
96. $\iiint_R (x+y+z) dx dy dz$, where $R: 0 \leq x \leq 1, 1 \leq y \leq 2, 2 \leq z \leq 3$ a. 4.5 b. 3.5 c. 2.5 d. 1.5
97. Evaluate the integral: $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$ a. $\frac{8}{3} \log 2 - \frac{19}{9}$ b. $\frac{8}{3} \log 2 + \frac{19}{9}$ c. $-\frac{8}{3} \log 2 + \frac{19}{9}$ d. $-\frac{8}{3} \log 2 - \frac{19}{9}$

98. $\int_0^{\log 2} \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz$ a. 0.625 b. 0.125 c. 0.375 d. NOTA

99. $\int_0^4 \int_0^x \int_0^{x+y} z dz dy dx$ a. 50 b. 60 c. 70 d. 80

100. $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x - y + z) dx dy dz$ a. 0 b. 1 c. 2 d. 4

Engr. almighty Formula